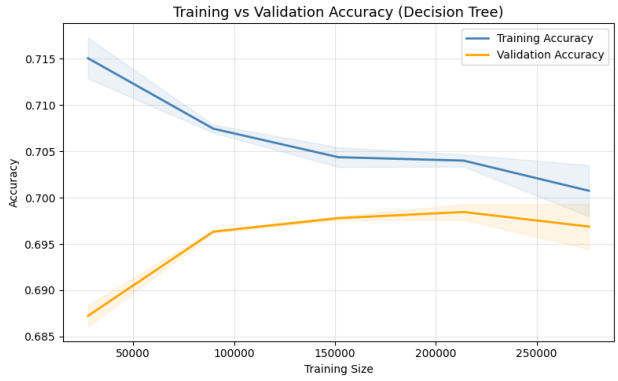
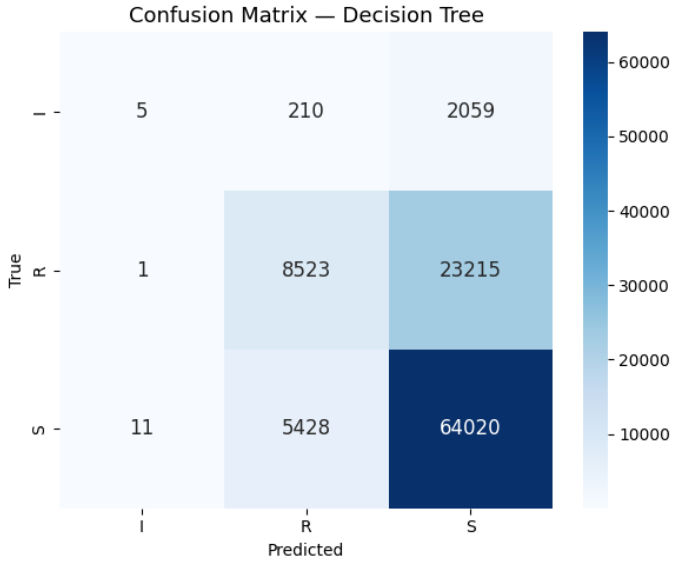
Decision Tree

ชื่อยา + Specimen + เพศ + อายุ + หอผู้ป่วย



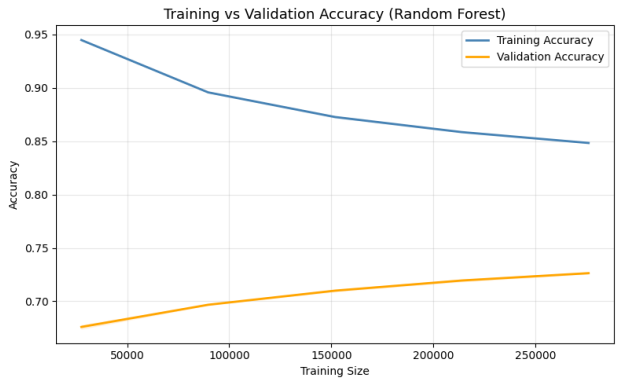


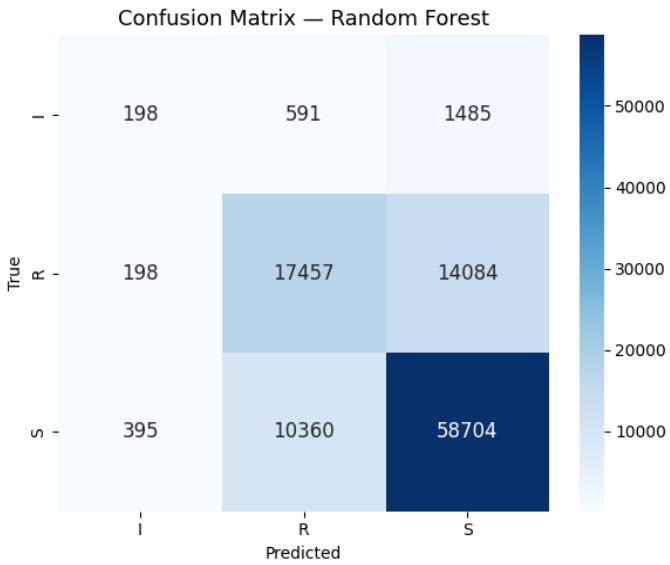
Classification Report:

Accuracy: 70.11%

	precision	recall	f1-score	support
I	0.29	0.00	0.00	2274
R	0.60	0.27	0.37	31739
S	0.72	0.92	0.81	69459
accuracy			0.70	103472
macro avg	0.54	0.40	0.39	103472
weighted avg	0.67	0.70	0.66	103472

Random Forest



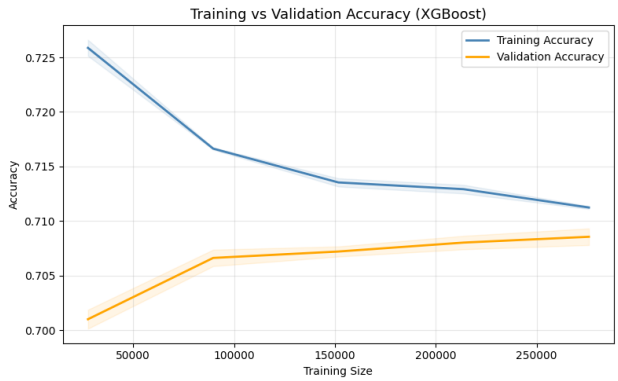


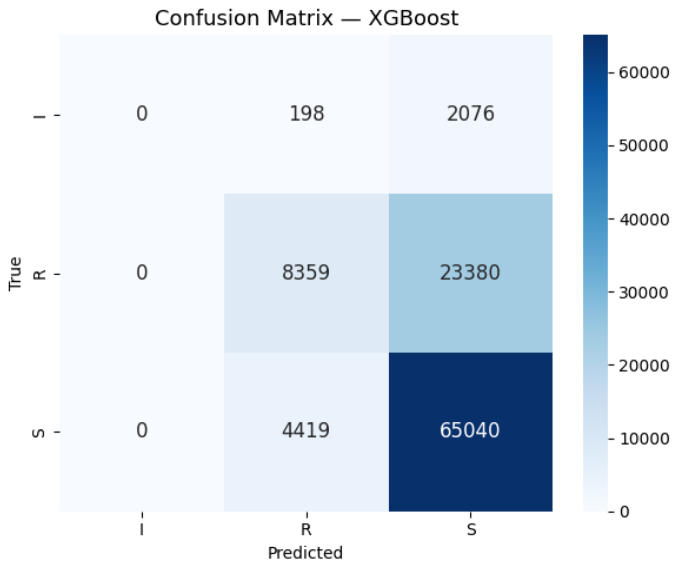
Classification Report:

Accuracy: 73.80%

	precision	recall	f1-score	support
I	0.25	0.09	0.13	2274
R	0.61	0.55	0.58	31739
S	0.79	0.85	0.82	69459
accuracy			0.74	103472
macro avg	0.55	0.49	0.51	103472
weighted avg	0.72	0.74	0.73	103472

XGBoost



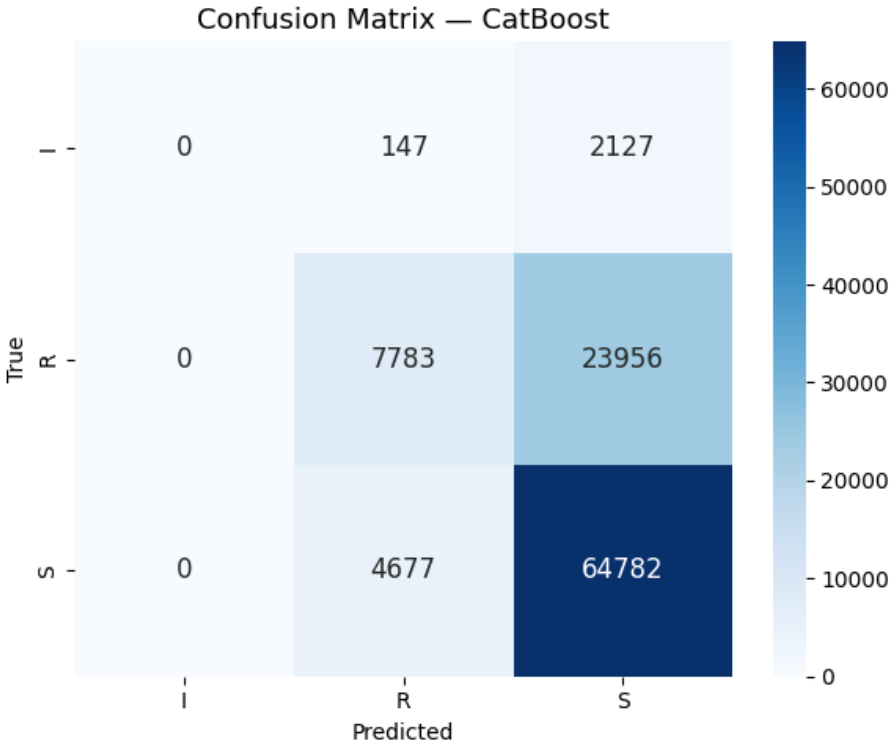


Classification Report:

Accuracy: 70.94%

	precision	recall	f1-score	support
I	0.00	0.00	0.00	2274
R	0.64	0.26	0.37	31739
S	0.72	0.94	0.81	69459
accuracy			0.71	103472
macro avg	0.45	0.40	0.40	103472
weighted avg	0.68	0.71	0.66	103472

CatBoost



Classification Report:

Accuracy: 70.13%

	precision	recall	f1-score	support
I	0.00	0.00	0.00	2274
R	0.62	0.25	0.35	31739
S	0.71	0.93	0.81	69459
accuracy			0.70	103472
macro avg	0.44	0.39	0.39	103472
weighted avg	0.67	0.70	0.65	103472

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STEP 6.7 — Model Comparison

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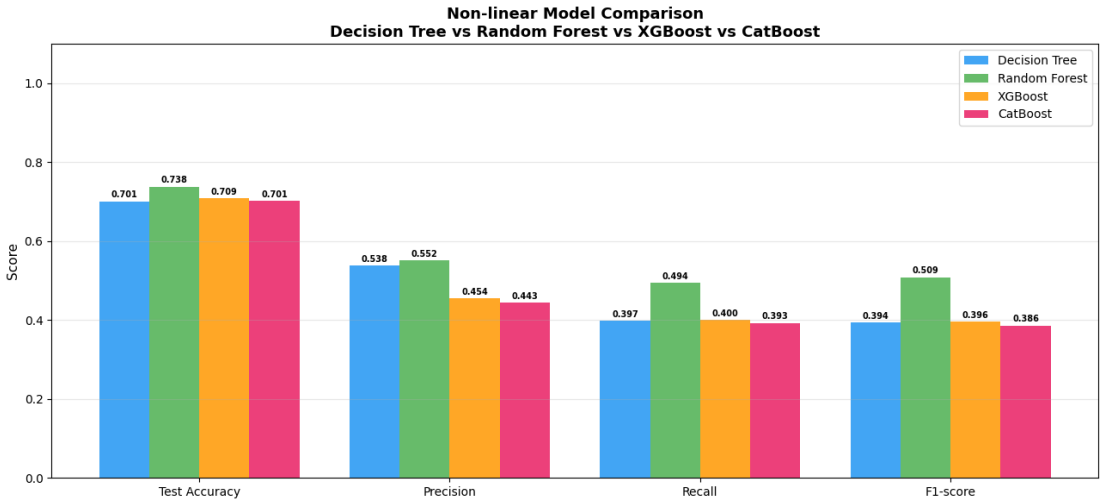
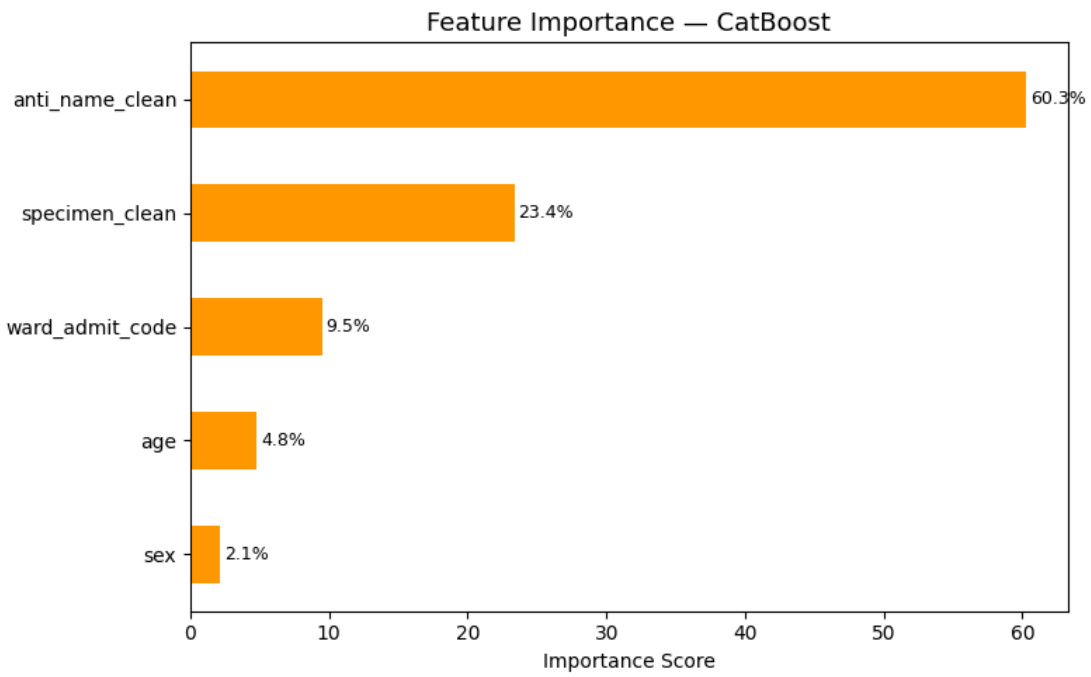
Model	Test Acc	Precision	Recall	F1
Decision Tree	0.7011	0.5376	0.3975	0.3941
Random Forest	0.7380	0.5517	0.4941	0.5088 ✅
XGBoost	0.7094	0.4543	0.3999	0.3957
CatBoost	0.7013	0.4434	0.3926	0.3864





Best Model : Random Forest

Test Accuracy : 73.80%

Macro F1 : 0.5088

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